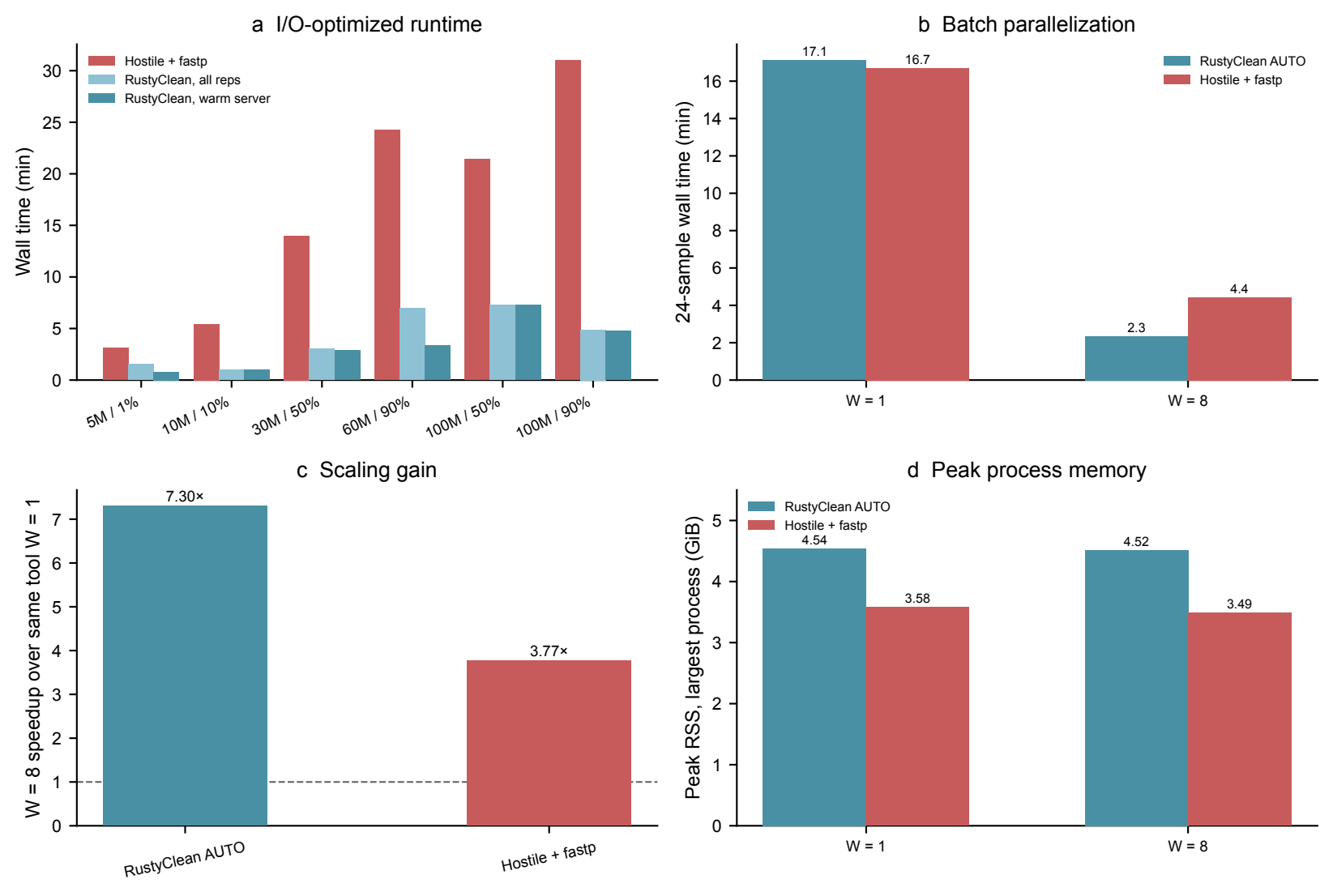
a I/O-optimized runtime
Hostile + fastp
RustyClean, all reps
RustyClean, warm server
Wall time (min)
0
5
10
15
20
25
30
5M / 1%
10M / 10%
30M / 50%
60M / 90%
100M / 50%
100M / 90%
b Batch parallelization
RustyClean AUTO
Hostile + fastp
24-sample wall time (min)
0
2
4
6
8
10
12
14
16
17.1
16.7
2.3
4.4
W = 1
W = 8
c Scaling gain
W = 8 speedup over same tool W = 1
0
1
2
3
4
5
6
7
7.30×
3.77×
RustyClean AUTO
Hostile + fastp
d Peak process memory
RustyClean AUTO
Hostile + fastp
Peak RSS, largest process (GiB)
0
1
2
3
4
5
4.54
3.58
4.52
3.49
W = 1
W = 8